Local projections: Identification, inference, variance decompositions

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Introduction

- My slides emphasize hands-on coding tools and intuitive concepts rather than math.
- I prioritize simplicity of implementation in Stata over cutting-edge econometric performance (but will give references to cutting-edge papers).
- I have uploaded Stata do-files to accompany the lectures. I encourage you to play around with the code and data.
- The slides cite the literature only sparingly.
- Reading list uploaded separately.

Outline

1. Identification by timing restrictions
2. Inference
3. Variance decompositions
4. Proxies, instruments
5. Other identification schemes

Local projection with control variables

- Local projection:

$$y_{t+h} = \mu_h + \theta_h x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}.$$

  - y_{t+h}: outcome variable
  - x_t: impulse variable
  - r_t: contemporaneous control variables (vector)
  - $w_{t-1}, \ldots, w_{t-p}$: lagged control variables (vector)
  - $\xi_{h,t+h}$: regression error

- The choice of y and x is determined by our economic hypothesis. But how should we choose which variables to control for? And how many lags?

Running example: Bilal & Känzig (2025) – henceforth BK

THE MACROECONOMIC IMPACT OF CLIMATE CHANGE:
GLOBAL VS. LOCAL TEMPERATURE

Adrien Bilal
Diego R. Känzig

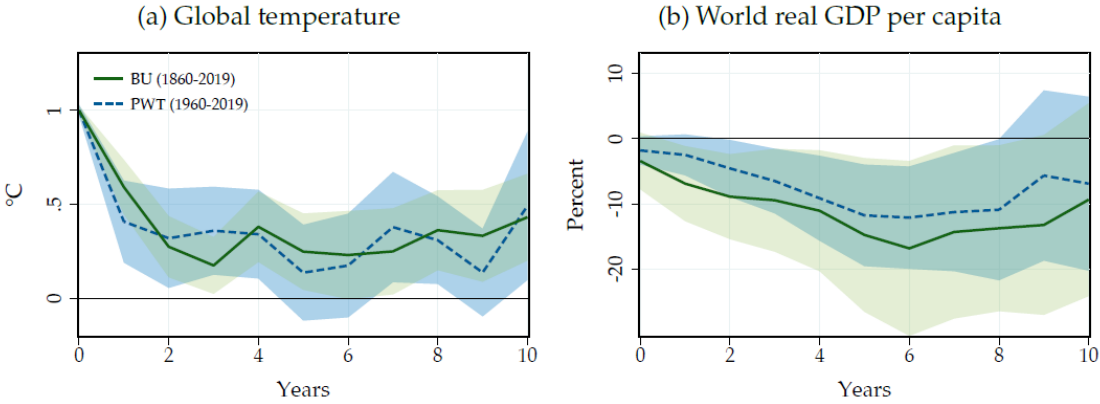
Working Paper 32450
http://www.nber.org/papers/w32450

Data: `https://github.com/dkaenzig/micc_replication`

Running example (BK)

- Study the effect of global temperature shocks on global GDP (among other things).
- Outcome y: global GDP.
- Impulse x: temperature anomaly = deviation of global temperature from trend.
  - Trend measured using Hamilton (2018) filter.
- Controls: recession indicator, oil price, 1-year U.S. treasury yield
- *Note:* My regression specifications differ slightly from those in the paper.

Figure 3: The Effect of Global Temperature Shocks on Temperature and World Output



Notes: Impulse responses of global mean temperature and world real GDP per capita to a global temperature shock, estimated based on (2). Controls: Lags of dependent variable, temperature shock, world real GDP growth, current and lagged recession dummies, and linear time trend. Lag length differs by sample (2 lags in PWT; 4 lags in BU). Lines: point estimate in PWT (dashed) and BU sample (solid). Shaded areas: 95% confidence bands for PWT (darker) and BU sample (lighter).

Source: Bilal & Känzig (2025)

```
. describe

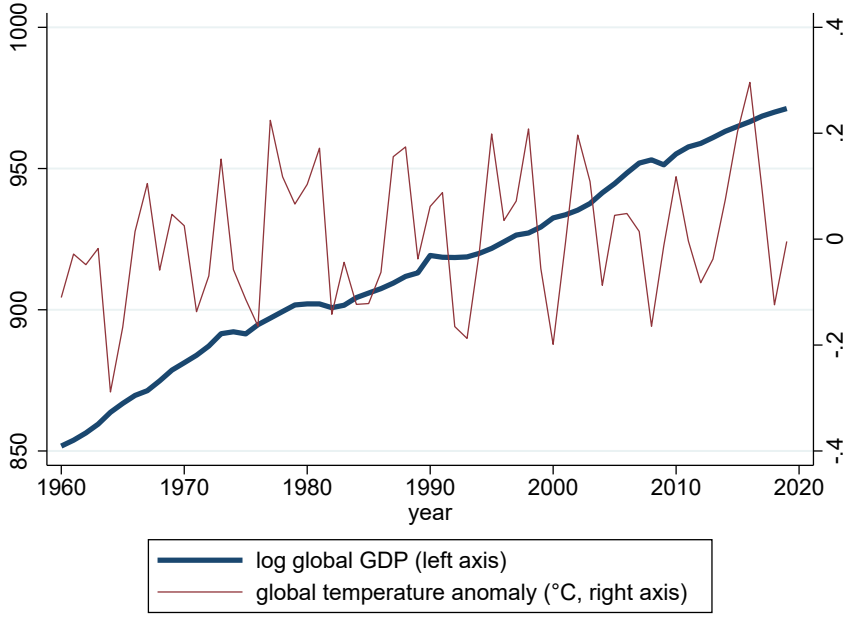
Contains data from micc_data.dta
 Observations:            60
    Variables:            12                  16 Nov 2024 09:25
                                              (_dta has notes)
```

Variable name	Storage type	Display format	Value label	Variable label
year	double	%10.0g		year
treasury1y	double	%10.0g		1Y US treasury yield
lnrgdppc_worl~t	float	%9.0g		log world real gdp per capita (national accounts), pwt
lnrgdppc_worl~i	float	%9.0g		log world real gdp per capita, wdi
lnpoil_wti	float	%9.0g		log WTI crude oil price
gtmp_noaa_aw	double	%14.2f		global absolute temperature, noaa
gtmp_bkly_aw	double	%14.2f		global absolute temperature, berkeley, constructed
gtmp_noaa_aw~1s	double	%10.0g		Forecast-error h=1, p=2, global temperature anomalies, noaa
gtmp_bkly_aw~1s	double	%10.0g		Forecast-error h=1, p=2, global temperature anomalies, berkeley, constructed
gtmp_noaa_aw~2s	double	%10.0g		Hamilton FE h=2, p=2, global temperature anomalies, noaa
gtmp_bkly_aw~2s	double	%10.0g		Hamilton FE h=2, p=2, global temperature anomalies, berkeley, constructed
recessiondates	float	%9.0g		Major global recession dates indicator

```
Sorted by: year
```

```
. summarize
```

Variable	Obs	Mean	Std. dev.	Min	Max
year	60	1989.5	17.46425	1960	2019
treasury1y	60	5.040667	3.314078	.12	14.78
lnrgdppc_w~t	60	915.2686	33.4173	851.7413	971.2475
lnrgdppc_w~i	60	879.9434	29.88987	819.0944	930.0361
lnpoil_wti	60	290.923	113.2949	107.1584	460.0841
gtmp_noaa_aw	60	14.26117	.3045034	13.75	14.93
gtmp_bkly_aw	60	14.47249	.3228001	13.90407	15.16037
gtmp_noaa~1s	60	.002426	.1056351	-.2456519	.209125
gtmp_bkly~1s	60	.0033159	.1168523	-.2792108	.2700942
gtmp_noaa~2s	60	.0031591	.1174761	-.2427712	.2844282
gtmp_bkly~2s	60	.0045739	.1255221	-.2885343	.2960396
recessiond~s	60	.2666667	.4459485	0	1


1000
950
900
850
.4
.2
0
-.2
-.4
1960
1970
1980
1990
2000
2010
2020
year
log global GDP (left axis)
global temperature anomaly (°C, right axis)

```
. * LP settings
. local outcome lnrgdppc_world_pwt

. local impulse gtmp_bkly_aw_dtfe2s

. local controls L(0/2).(recessiondates) L(1/2).(`outcome' `impulse' lnpoil_wti treasury1y)

. local maxhorz = 10
```

```
. local sample "1962,2015"

. reg F4.`outcome' `impulse' `controls' if tin(`sample')
```

Source	SS	df	MS	Number of obs	=	54
				F(12, 41)	=	553.08
Model	44083.7203	12	3673.64336	Prob > F	=	0.0000
Residual	272.330796	41	6.64221454	R-squared	=	0.9939
				Adj R-squared	=	0.9921
Total	44356.0511	53	836.906624	Root MSE	=	2.5772

F4. lnrgdppc_world_pwt	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
gtmp_bkly_aw_dtfe2s	-6.729412	3.6077	-1.87	0.069	-14.01531	.5564857
recessiondates						
--.	-1.800983	1.125927	-1.60	0.117	-4.074839	.4728735
L1.	-1.567738	1.212745	-1.29	0.203	-4.016925	.8814499
L2.	-1.731939	1.231123	-1.41	0.167	-4.218243	.7543646
lnrgdppc_world_pwt						
L1.	.498831	.3512466	1.42	0.163	-.2105259	1.208188
L2.	.4788941	.3501463	1.37	0.179	-.2282407	1.186029
gtmp_bkly_aw_dtfe2s						
L1.	-2.148618	3.554991	-0.60	0.549	-9.328067	5.030831
L2.	-3.144832	3.96723	-0.79	0.433	-11.15682	4.867152
lnpoil_wti						
L1.	.0115704	.0206165	0.56	0.578	-.0300655	.0532063
L2.	-.0160061	.0187146	-0.86	0.397	-.0538009	.0217888
treasury1y						
L1.	-.7379184	.3328448	-2.22	0.032	-1.410112	-.0657246
L2.	.3279273	.327486	1.00	0.323	-.3334441	.9892987
_cons	36.18253	38.37409	0.94	0.351	-41.31552	113.6806

This is the LP at horizon $h = 4$.

A 1°C temperature anomaly is associated with 6.7% lower global real GDP after 4 years.

What is a local projection estimating?

$$y_{t+h} = \mu_h + \theta_h x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}$$

- Frisch-Waugh ("residual regression") Theorem: instead of running the above multiple regression, we can equivalently compute θ_h in two steps.
  1. Regress x_t on the controls $r_t, w_{t-1}, \ldots, w_{t-p}$. Compute residual $\tilde{x}_t$.
  2. Regress y_{t+h} on $\tilde{x}_t$ (with no controls).
- Interpretation: an LP is measuring the response of y_{t+h} to 1-unit impulse in the shock $\tilde{x}_t$.
- For our LP results to have meaning, we need to pick control variables so that the implied shock $\tilde{x}_t$ is an economically sensible quantity.

Running example (BK): interpretation of the shock

- In the BK data, $\tilde{x}_t$ = movements in temperature that cannot be explained by...
  - current or lagged global recessions,
  - lagged global temperature,
  - lagged world GDP,
  - lagged oil prices,
  - lagged interest rates.
- This removes the endogenous feedback from past economic conditions to current temperature (e.g., through emissions).

```
. reg `impulse' `controls' if tin(`sample')
```

Source	SS	df	MS	Number of obs	=	54
				F(11, 42)	=	2.21
Model	.295745625	11	.026885966	Prob > F	=	0.0321
Residual	.510331214	42	.012150743	R-squared	=	0.3669
				Adj R-squared	=	0.2011
Total	.806076839	53	.015208997	Root MSE	=	.11023

gtmp_bkly_aw_dtfe2s	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
recessiondates						
--.	.0527202	.0474645	1.11	0.273	-.043067	.1485074
L1.	-.103721	.0493389	-2.10	0.042	-.2032909	-.004151
L2.	-.0512201	.0520593	-0.98	0.331	-.1562801	.0538398
lnrgdppc_world_pwt						
L1.	-.0119103	.0149102	-0.80	0.429	-.0420002	.0181797
L2.	.0132207	.0148364	0.89	0.378	-.0167203	.0431617
gtmp_bkly_aw_dtfe2s						
L1.	.1371779	.1505684	0.91	0.367	-.1666814	.4410372
L2.	-.5108355	.150261	-3.40	0.001	-.8140744	-.2075965
lnpoil_wti						
L1.	.000578	.0008773	0.66	0.514	-.0011924	.0023484
L2.	-.0004909	.0007968	-0.62	0.541	-.002099	.0011171
treasury1y						
L1.	.0106429	.0141409	0.75	0.456	-.0178946	.0391804
L2.	-.0041879	.0139918	-0.30	0.766	-.0324246	.0240488
_cons	-1.19898	1.630821	-0.74	0.466	-4.490111	2.09215

```
. predict impulse_resid if tin(`sample'), resid
(6 missing values generated)
```

```
. reg F4.`outcome' impulse_resid if tin(`sample')
```

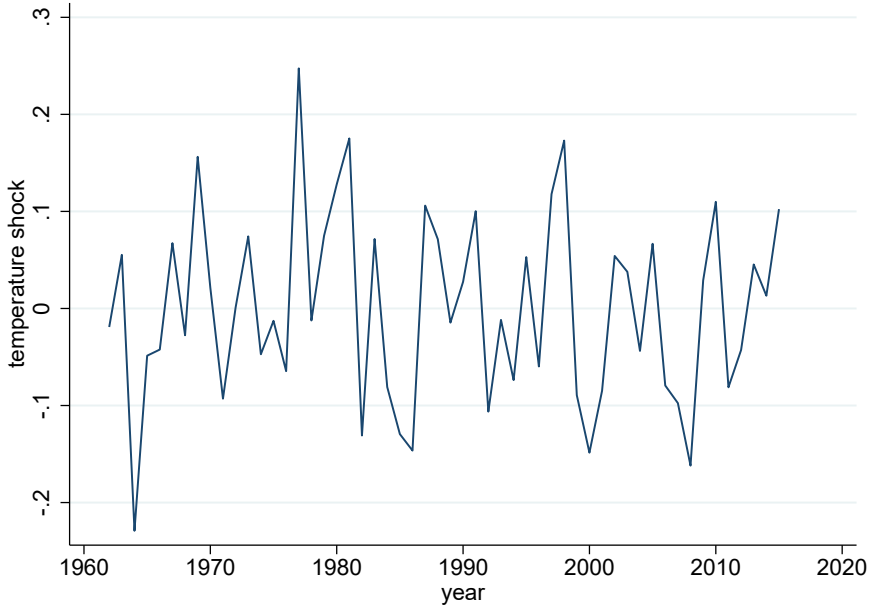
Source	SS	df	MS	Number of obs	=	54
				F(1, 52)	=	0.03
Model	23.1103465	1	23.1103465	Prob > F	=	0.8699
Residual	44332.9407	52	852.556553	R-squared	=	0.0005
				Adj R-squared	=	-0.0187
Total	44356.0511	53	836.906624	Root MSE	=	29.199

F4. lnrgdppc_wo~t	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
impulse_resid	-6.729413	40.87291	-0.16	0.870	-88.7469	75.28808
_cons	921.558	3.973422	231.93	0.000	913.5848	929.5313

```
. su impulse_resid, detail
```

Residuals

	Percentiles	Smallest		
1%	-.229061	-.229061		
5%	-.148569	-.1619047		
10%	-.1294395	-.148569	Obs	54
25%	-.0793732	-.1464162	Sum of wgt.	54
50%	-.0120584		Mean	-1.72e-10
		Largest	Std. dev.	.0981269
75%	.0714048	.1562584		
90%	.1178279	.1729615	Variance	.0096289
95%	.1729615	.1752104	Skewness	.1287115
99%	.2474069	.2474069	Kurtosis	2.671895


temperature shock
.3
.2
.1
0
-.1
-.2
1960
1970
1980
1990
2000
2010
2020
year

Properties of a shock

- For the implicit $\tilde{x}_t$ to be a meaningful economic shock, it should be...
  1. serially uncorrelated,
  2. unpredictable by other lagged time series,
  3. able to be mapped to a theoretical counterpart in a (hypothetical) DSGE model.
- If #1 fails, we can include additional lags in our LP specification (more soon).
- If #2 fails, we can control for additional variables in our LP. But we should also question whether we have thought through all possible channels of reverse causality.
  - Example (BK): if past natural disasters predict temperature shocks, maybe we're not fully taking into account the endogeneity of the global climate system.
- If #3 fails, need to think harder about the purpose of running the LP in the first place.
  - Example: if x = wage, what is the theoretical counterpart of this supposed "wage shock"?

Discussion: implicit LP shock

- Just because the Frisch-Waugh Theorem says we *can* compute the LP estimate in two steps doesn't mean we *should* in practice. In particular, the standard errors from the second step should not be used.
- Best to run a single-step LP regression with all controls. Then it's easy to compute standard errors, as we'll see.
- But it's useful to compute the implicit residualized shock, plot it, and report summary statistics. For example, its standard deviation informs us about the typical scale of the shock.
  - Std. dev. $\approx 0.1^\circ$C in the running example (BK). So if we report impulse responses with respect to a 1° increase in the temperature anomaly x_t, that corresponds to an unusually large shock.

Timing restrictions: contemporaneous and lagged controls

$$y_{t+h} = \mu_h + \theta_h x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}$$

- Identification with control variables can be made concrete using the framework of short-run timing restrictions, just like a recursively/Cholesky identified SVAR model.
- Suppose we hope to identify the impulse response with respect to some underlying, unobserved shock ε_t (e.g., a climate shock).
- Contemporaneous controls r_t should not respond *within the period* to ε_t.
- The impulse variable x_t can respond to r_t and to ε_t, but not to any other underlying shocks *within the period*.
- If these conditions fail, then the contemporaneous controls "over-control", since they are removing part of the effect of ε_t.

Running example (BK): identification assumptions

- In the BK specification, they control for one contemporaneous regressor: a global recession indicator.
- Implicit assumption 1: an unexpected climate shock can cause a global recession, but only with a lag of at least one year.
- Implicit assumption 2: once we control for shocks that immediately cause global recessions, any left-over high-frequency movement in the temperature anomaly is due to a well-defined climate shock.
  - There can't be any other kind of economic shock that affects temperature anomalies in the short run.
  - In particular, there can't be measurement error in the temperature anomaly series.

Discussion: selection of control variables

- The choice of control variables should be motivated by economic theory.
  - What are possible channels of reverse causality (i.e., y causing x)?
  - What timing assumptions are we willing to make, if any? Same as in SVAR analysis.
- If in doubt, include more lagged controls.
  - Even if not needed for identification, they can soak up residual variance and thereby decrease standard errors (if sample size is not too small).
- If a potential control variable is insignificant in *both* the LP outcome equation *and* in the regression of x_t on all controls, then it can be dropped, but not otherwise.
  - Corollary: always control at least for lags of y and lags of x.

Lag length selection

- Economic theory rarely provides guidance on exactly how many lags to control for.
- Data-based lag selection using an auxiliary VAR:
  1. Consider VAR specification containing all the variables in the analysis: y, x, r, w.
  2. Use information criteria to select the VAR lag length $\hat{p}$. AIC recommended.
  3. Throw away auxiliary VAR results (only use it to select $\hat{p}$).
  4. Run LP that controls for $\hat{p}$ lags of all the data series.
- If in doubt, include more lags.

Running example (BK): lag length selection

```
. varsoc `outcome' `impulse' recessiondates lnpoil_wti treasury1y, maxlag(4)

Lag-order selection criteria

  Sample: 1964 thru 2019                                  Number of obs = 56
```

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-671.149				21137.7	24.1482	24.2183	24.329
1	-394.112	554.07	25	0.000	2.61586*	15.1469*	15.5675*	16.2319*
2	-370.66	46.903	25	0.005	2.82536	15.2022	15.9734	17.1913
3	-344.597	52.128	25	0.001	2.87648	15.1642	16.2859	18.0575
4	-319.689	49.815*	25	0.002	3.2232	15.1675	16.6398	18.965

```
  * optimal lag
  Endogenous: lnrgdppc_world_pwt gtmp_bkly_aw_dtfe2s recessiondates
              lnpoil_wti treasury1y
   Exogenous: _cons
```

Here AIC selects 1 lag. BK control for 2 lags in their LPs, which is a safe choice.

LP with observed shock

- In some applications, researchers observe a variable x_t that can directly be viewed as an unpredictable shock.
- Example: Känzig (2021) measures changes in oil futures prices in short time windows around OPEC announcements.
  - If financial markets are efficient, these changes should be unpredictable.

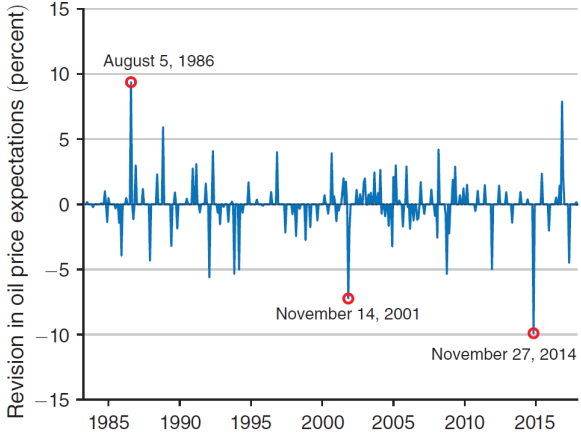


FIGURE 1. THE OIL SUPPLY SURPRISE SERIES

Note: This figure shows the oil supply surprise series, constructed as the first principal component from changes in oil futures prices (using the 1-month to 12-month WTI crude contracts) around OPEC announcements, scaled to match the average volatility of the underlying price changes.

Source: Känzig (2021)

LP with observed shock

- If we already observe the shock we care about, then control variables are technically not needed for identification.
- A LP of an outcome y on the observed shock x consistently estimates the impulse responses.
- But even in this case, we should still control for lagged variables, such as lagged y and x. This decreases the standard errors and makes inference simpler (more later).
  - If point estimates change by a lot, then the "shock" is actually predictable...
  - Don't include contemporaneous controls (unless you want to impose an additional timing restriction, but this is not needed for consistency).

Outline

1. Identification by timing restrictions
2. Inference
3. Variance decompositions
4. Proxies, instruments
5. Other identification schemes

Inference on impulse responses

- We would like to quantify the uncertainty in the estimated impulse responses. This is typically substantial in time series data.
- We would also like to test hypotheses involving the impulse response parameters.
- To do this, we need standard errors (SE) for the LP regression coefficient on x.
- We cannot blindly rely on Stata commands for cross-sectional regression, since our data is serially correlated.

Serially correlated errors

$$y_{t+h} = \mu_h + \theta_h x_t + \gamma_h' r_t + \delta_{h,1}' w_{t-1} + \cdots + \delta_{h,p}' w_{t-p} + \xi_{h,t+h}$$

- The LP error term $\xi_{h,t+h}$ is typically serially correlated, since it's a multi-step forecast error that should be a linear combination of shocks that hit between time t and $t+h$.
- Example: AR(1) model.

$$y_t = \rho y_{t-1} + \varepsilon_t \quad \Longrightarrow \quad y_{t+2} = \rho^2 y_t + \underbrace{\varepsilon_{t+2} + \rho \varepsilon_{t+1}}_{\xi_{2,t+2}}.$$

- In general, we need to correct standard errors for the serial correlation in this error term (HAC: Heteroskedasticity and Autocorrelation Consistent SE).

Serially correlated errors

- The standard way to do HAC corrections is to use the Newey-West (NW) variance estimator.
- This requires a choice of bandwidth, which determines the longest-lag autocovariance of the data that is included in the correction.
- In Stata, the `ivreg2` command can compute OLS regressions with NW standard errors, using the automatic bandwidth selection procedure of Newey & West (1994).
- However, Newey-West SE are known to be downward-biased in small samples, often severely.
  - See Lazarus, Lewis, Stock & Watson (2018) for updated recommendations on HAC inference. Requires different variance estimators, bandwidth choices, and/or critical values.

Simpler standard errors for LP

- Luckily, the usual (cross-sectional) heteroskedasticity-robust standard errors suffice for LP, provided we control for lags of the data ("lag augmentation").
- We can then just use the standard `regress ..., robust` command in Stata.
  - No need to choose any bandwidth.
- Since we already know we should control for lagged data for reasons of identification and efficiency, the simpler SE procedure is a nice bonus.
- For the simple SE to be valid, we need:
  1. To control for sufficient lags (for example chosen by AIC).
  2. Technical condition: shock of interest is nonlinearly unpredictable by *future* shocks. Holds for many shock processes, e.g., GARCH. Even when assumption fails, the simpler SE may work better in small samples than NW SE (Herbst & Johannsen, 2024).

Running example (BK): estimates and SE in Stata

```
postfile f h ir se se_nw using lp, replace
forvalues h=0/`maxhorz' {
  * LP with Newey-West SE
  qui ivreg2 F`h'.`outcome' `impulse' `controls', robust kernel(bartlett) bw(auto)
  local nw_se = _se["`impulse'"]
  * LP with heteroskedasticity-robust SE
  qui reg F`h'.`outcome' `impulse' `controls', robust
  * Store horizon, coefficient, and SE
  post f (`h') (_b["`impulse'"]) (_se["`impulse'"]) (`nw_se')
}
postclose f
```

Note: Stata (version 18 and newer) has a purpose-built command for local projections: `lpirf`. But I find its syntax confusing and so will avoid it.

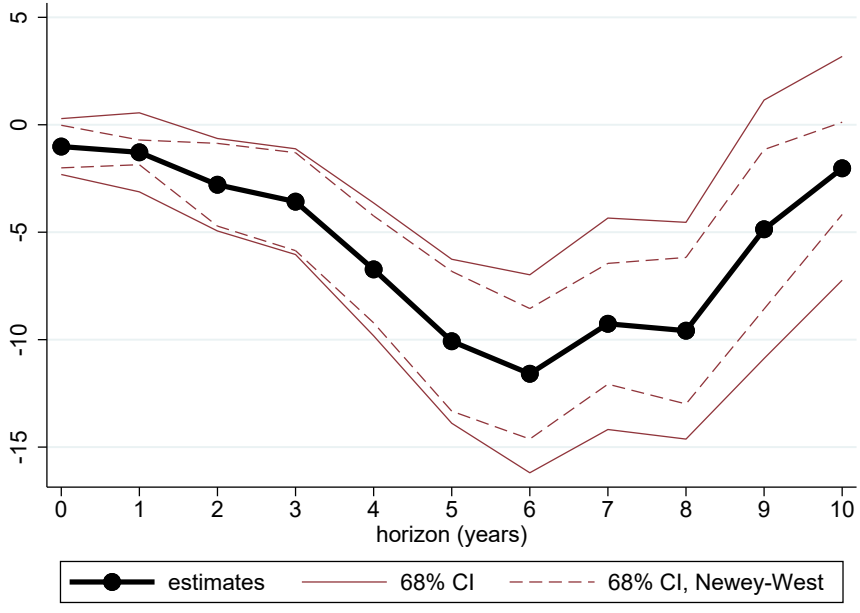
```
. use lp, clear

. list h ir se se_nw
```

	h	ir	se	se_nw
1.	0	-1.01357	1.300066	.9911674
2.	1	-1.282557	1.838438	.5748452
3.	2	-2.790329	2.15099	1.924042
4.	3	-3.577711	2.459928	2.279449
5.	4	-6.729412	3.091988	2.487145
6.	5	-10.07387	3.815386	3.252955
7.	6	-11.58745	4.606543	3.036841
8.	7	-9.260947	4.919593	2.810547
9.	8	-9.583155	5.043317	3.408893
10.	9	-4.863534	6.013761	3.710221
11.	10	-2.026347	5.206398	2.147646

Confidence band

- Since the LP estimator is asymptotically normally distributed, we can construct a...
  - 68% confidence band: `estimate` $\pm$ `SE`,
  - 90% confidence band: `estimate` $\pm$ `1.64 * SE`,
  - 95% confidence band: `estimate` $\pm$ `1.96 * SE`.
- The 68% band, say, has the following property (in large samples): for each impulse response horizon *separately*, the band (i.e., interval) has a 68% probability of covering the true impulse response.
- Because the bands only guarantee a certain coverage probability *for each horizon separately*, they cannot be used to draw inferences about hypotheses that involve multiple horizons, e.g., whether the IRF is monotonic. (More later.)


5
0
-5
-10
-15
0
1
2
3
4
5
6
7
8
9
10
horizon (years)
estimates
68% CI
68% CI, Newey-West

Discussion: standard errors

- I would recommend using simple heteroskedasticity-robust SE over Newey-West.
  - But important to control for lagged data, guided by AIC, as previously discussed.
- More advanced HAC procedures could be used, but requires choices of tuning parameters such as bandwidth (Lazarus et al., 2018).
- Xu (2023) provides a SE formula for LP that works more generally than the simple SE and is also free of tuning parameters. Useful as a robustness check.
- When the data is very persistent, we will see later that the simple LP confidence band can be improved using a bootstrap.

Outline

1. Identification by timing restrictions
2. Inference
3. Variance decompositions
4. Proxies, instruments
5. Other identification schemes

Forecast error variance decomposition

- We would like to quantify how important our (implicit) shock $\tilde{x}_t$ is for the outcome y_t.
  - Recall: $\tilde{x}_t$ = residual in regression of x_t on all the controls $r_t, w_{t-1}, \ldots, w_{t-p}$.
- Forecast Error Variance Decomposition (FEVD): How much would our forecast error for y_{t+h} decrease if we knew the intervening shocks $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$, compared to if we didn't?
- Gorodnichenko & Lee (2020) procedure (separately for each horizon h):
  1. Regress y_{t+h} on all controls variables $r_t, w_{t-1}, \ldots, w_{t-p}$. Compute residual $\hat{f}_{h,t+h}$.
  2. $FEVD_h = R^2$ from regression of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \tilde{x}_{t+1}, \ldots, \tilde{x}_{t+h}$.
- *Note:* This definition depends on the baseline set of controls. We're asking how well current+future $\tilde{x}_t$ explain the forecast error from the regression with baseline controls.

Running example (BK): forecast error variance decomposition

```
reg ‘impulse’ ‘controls’
predict impulse_resid_full, resid // Estimate shock on full available sample
postfile f h r2 using fevd, replace
forvalues h=0/‘maxhorz’ {
  * Regression w/o shock
  qui reg F‘h’.‘outcome’ ‘controls’
  qui predict res, resid // Residual
  * Regress residual on current+future shocks
  qui reg res F(0/‘h’).impulse_resid_full
  * Store horizon and R^2
  post f (‘h’) (e(r2))
  drop res
}
postclose f
```

```
. use fevd, clear

. list h r2
```

	h	r2
1.	0	.0082018
2.	1	.026708
3.	2	.0851245
4.	3	.105231
5.	4	.1845562
6.	5	.3068271
7.	6	.4491798
8.	7	.4915972
9.	8	.4603188
10.	9	.4598588
11.	10	.4383282

At horizon $h = 2$ years, climate shocks explain 8.5% of the variance of the GDP forecast error.

At $h = 7$ years, they explain 49.2%!

Outline

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Proxies

- Suppose we have constructed a direct measure z_t of an underlying shock ε_t, but it's contaminated with measurement error v_t:

$$z_t = \alpha + \beta\varepsilon_t + v_t.$$

  We call such a z_t a proxy or external instrument for ε_t.

  - Example: maybe Känzig's (2021) oil news shock is contaminated by financial market microstructure noise.

- If the measurement error v_t is classical (uncorrelated with *any* systematic economic shocks), then an LP regression of y_{t+h} on z_t and controls will suffer from attenuation bias: the impulse response will be biased toward 0.

  - This is because a 1-unit increase in z_t does not correspond to a 1-unit increase in the unobserved shock ε_t – there is also noise.

Proxies: unit effect normalization

- Problem: we cannot measure the effect of a 1-unit increase in the unobserved ε_t.
- Solution: report the effect of an impulse that causes a 1-unit increase in some observed variable s_t (the normalization variable), on impact.
- Let $\theta_{z \to y,h}$ be the coefficient from an LP of y_{t+h} on z_t.
- Let $\theta_{z \to s,h}$ be the coefficient from an LP of s_{t+h} on z_t.
- Then the ratio

$$\frac{\theta_{z \to y,h}}{\theta_{z \to s,0}}$$

measures the response of y_{t+h} from an impulse in z_t of a magnitude such that it raises s_t by 1 unit on impact. This is a unit-effect-normalized impulse response.

Proxies: unit effect normalization

$$\frac{\theta_{z \to y,h}}{\theta_{z \to s,0}}$$

- Example (Känzig, 2021):
  - y: U.S. GDP.
  - z: high-frequency change in oil futures around OPEC announcements.
  - s: spot oil price ($).
- The unit-effect-normalized impulse response measures the effect after h time periods of an impulse in oil futures on U.S. GDP, for an impulse of a magnitude that raises the spot price by 1$ on impact.

Proxies: unit effect normalization

$$\frac{\theta_{z\to y,h}}{\theta_{z\to s,0}}$$

- We've already established that both the numerator and denominator suffer from attenuation bias, due to the noise in z:

  $$\theta_{z\to y,h} = \text{bias} \times \theta_{\varepsilon\to y,h}, \quad \theta_{z\to s,0} = \text{bias} \times \theta_{\varepsilon\to s,0},$$

  where "bias" is a number somewhere between 0 and 1.

- But this bias factor is the *same* in both the numerator and denominator, since it's the same proxy z.

- So the bias gets canceled out in the unit effect normalization:

  $$\frac{\theta_{z\to y,h}}{\theta_{z\to s,0}} = \frac{\theta_{\varepsilon\to y,h}}{\theta_{\varepsilon\to s,0}}.$$

LP-IV

$$\frac{\theta_{z\to y,h}}{\theta_{z\to s,0}} = \frac{\theta_{\varepsilon\to y,h}}{\theta_{\varepsilon\to s,0}}.$$

- We can therefore measure the causal effect of the underlying shock ε on y by running two LPs ($z \to y$ and $z \to s$) and taking a ratio. We're measuring the response to a shock ε of a magnitude such that it raises s by 1 unit on impact.

- But as you may remember from first-year econometrics class, this ratio is exactly the same as an instrumental variable (IV) regression

$$y_{t+h} = \mu_h + \tilde{\theta}_h s_t + \xi_{h,t+h} \quad \text{using } z_t \text{ as an IV.}$$

- This estimator $\tilde{\theta}_h$ of the unit-effect-normalized impulse response is called LP-IV.

LP-IV: control variables and SE

- We can allow the proxy to be correlated with lagged control variables:

$$z_t = \alpha + \beta\varepsilon_t + \psi_1' w_{t-1} + \cdots + \psi_p' w_{t-p} + v_t.$$

- Then we just need to control for $w_{t-1}, \ldots, w_{t-p}$ in our IV regressions.
- Controlling for lagged data has the same benefits as we emphasized earlier, so we should always control for (at least) lagged outcomes, proxies, and normalization variables.
- If we control sufficiently for lags, we can use the conventional heteroskedasticity-robust standard errors for IV regression.

Running example (BK): LP-IV

- It's reasonable to expect that the temperature anomalies are measured with error.
- We can estimate normalized impulse responses by LP-IV.
- Normalization variable = global temperature. So the scale of the climate shock is such that it raises global temperature by 1°C on impact.

```
. local norm gtmp_bkly_aw

. reg `norm' `impulse' `controls', robust
```

Linear regression

Number of obs	= 58
F(12, 45)	= 145.86
Prob > F	= 0.0000
R-squared	= 0.9637
Root MSE	= .06922

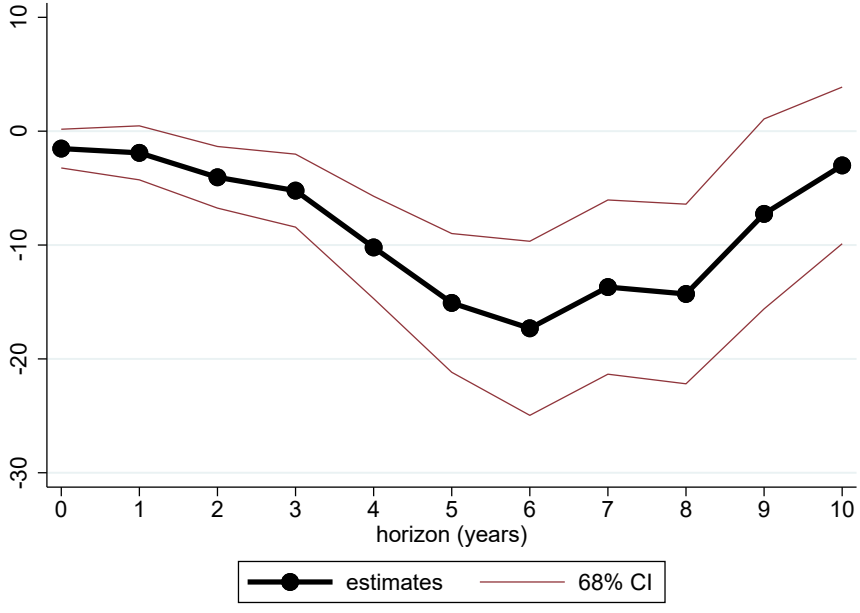
gtmp_bkly_aw	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
gtmp_bkly_aw_dtfe2s	.6629039	.0718191	9.23	0.000	.5182527	.807555
recessiondates						
--.	-.0185086	.0252853	-0.73	0.468	-.0694359	.0324187
L1.	-.0087899	.0286528	-0.31	0.760	-.0664996	.0489198
L2.	-.086643	.0293059	-2.96	0.005	-.1456681	-.0276178
lnrgdppc_world_pwt						
L1.	-.0071883	.0102486	-0.70	0.487	-.02783	.0134534
L2.	.0128437	.0095678	1.34	0.186	-.0064268	.0321142
gtmp_bkly_aw_dtfe2s						
L1.	-.2233564	.091459	-2.44	0.019	-.4075642	-.0391486
L2.	.3258549	.0969534	3.36	0.002	.1305808	.521129
lnpoil_wti						
L1.	.0005037	.0004839	1.04	0.304	-.000471	.0014783
L2.	.0002654	.0004162	0.64	0.527	-.000573	.0011037
treasury1y						
L1.	-.0086151	.0084964	-1.01	0.316	-.0257277	.0084976
L2.	-.0055693	.0083541	-0.67	0.508	-.0223953	.0112567

This is the first stage regression ($z \to s$).

It shows that global temperature rises by 0.66°C given a 1-unit impulse in the temperature anomaly.

So we need to scale the outcome LPs by $1/0.66$ – or just run LP-IV.

```
postfile f h ir se using lp_iv, replace
forvalues h=0/‘maxhorz’ {
  * IV regression
  qui ivreg2 F‘h’.‘outcome’ ‘controls’ (‘norm’ = ‘impulse’), robust
  * Store horizon, coefficient, and SE
  post f (‘h’) (_b["‘norm’"]) (_se["‘norm’"])
}
postclose f
```


10
0
-10
-20
-30
0
1
2
3
4
5
6
7
8
9
10
horizon (years)
estimates
68% CI

Proxy assumptions

- Let $\tilde{z}_t = \alpha + \beta\varepsilon_t + v_t$ be the residual from a regression of z_t on the lagged controls.
- Then consistency of the LP-IV estimator requires:
  1. $\beta \neq 0$: the proxy needs to be relevant for the shock of interest.
  2. v_t is classical measurement error: it's uncorrelated with *all* systematic economic shocks at all points in time, and therefore uncorrelated at all leads and lags with ε_t, y_t, s_t, and other macro variables.

Narrative restrictions

- One way to construct a proxy in practice is via qualitative narrative restrictions.
- Suppose we have information about a subset of historical dates where there were especially large *positive* or *negative* shocks of a certain kind (e.g., monetary policy shocks). At other dates we're unsure about which shocks hit the economy.
- Construct (noisy) proxy:

$$z_t = \begin{cases} +1 & \text{if large, positive shock at time } t, \\ -1 & \text{if large, negative shock at time } t, \\ 0 & \text{otherwise.} \end{cases}$$

✓ Positively correlated with the shock of interest.

? Uncorrelated with the other economic shocks if (i) these other shocks do not influence which dates we have information on, and (ii) misclassifications are random.

Weak instruments

- If the normalization variable s_t is only weakly correlated with the proxy z_t (conditional on controls), then we have a weak instrument, as the LP-IV estimator is nearly dividing by 0.
- This can lead to undesirable econometric properties: high (median-)bias and poor coverage of the conventional confidence band.
- There are two reasons why we could have weak instruments:
  1. The proxy z is dominated by noise, so only weakly correlated with the shock of interest ε.
  2. The normalization variable s responds only weakly to the shock of interest ε on impact.
- We should pick a normalization variable that correlates strongly with the proxy.
  - If this is not possible, we could construct weak-IV-robust confidence sets, but that's outside the scope of this workshop.

Proxy: variance decompositions

- If we only observe a proxy for the true shock, then the FEVD is not identified.
- The reason is that we do not know what share of the variance of the proxy is due to the shock, and what share is due to noise.
- If we treat the proxy as if it were an observed shock in the Gorodnichenko-Lee procedure, we *under-estimate* the FEVD due to the noise.
- Plagborg-Møller & Wolf (2022) show that we can also compute non-trivial *upper* bounds on variance decompositions.
  - Matlab code available, but not easy to implement directly in Stata.

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Other identification schemes

- LPs can also be used to estimate impulse responses with respect to shocks that are identified in more "exotic" ways:
  - Long-run restrictions, identif'n via heteroskedasticity or non-Gaussianity, etc.
- At a basic level, we could estimate the shock $\hat{\varepsilon}_t$ using an SVAR identification procedure, and then run an LP of y_{t+h} on $\hat{\varepsilon}_t$ (with lagged controls).
- But there also exist more direct implementations of many of these strategies in IV versions of LP (Plagborg-Møller & Wolf, 2021). Even for sign identification.
- Kolesár & Plagborg-Møller (2025) argue that some of these alternative identif'n schemes are less robust to nonlinearities in the DGP than LP with an observed shock/proxy.
  - If you can measure a shock directly (perhaps with measurement error), you should do it! Not only more transparent, but also more robust to nonlinearities.

What did we discuss?

- Selection of control variables + lag length, implicit shock.
- Simple vs. Newey-West SE, confidence band.
- FEVD.
- Proxies, unit effect normalization, LP-IV, narrative proxies.
- Other identification strategies: technically possible with LP, but better to directly measure a shock/proxy if feasible.